MIL-A-356A(OS)
25 September 1973
SUPERSEDING
JAN-A-356
18 June 1946

MILITARY SPECIFICATION

ASPHALTUM (GILSONITE)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one grade of powdered asphaltum. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-S-366	Sieve, Test
SS-R-406	Road and Paving Materials, Methods of Sampling and Testing

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9620

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3. REQUIREMENTS

3.1 **Material**. The material shall originate from natural deposits equal in quality to those of Utah and Colorado.

3.2 **Chemical and physical properties**. Asphaltum shall conform to the following requirements:

Specific gravity 25°/25° C	1.05 ± 0.05
Softening point, °C (min.)	125
Mineral matter, perc nt (max.)	1.0
Grit	None
Volatile matter, percent (max.)	1.0
Reaction of water extract	Alkaline to methyl orange
Solubility, percent (min.)	
in CCl_4	99
in petroleum ether	30
in CS_2	99

3.3 **Granulation**. Not less than 99.0 percent of the asphaltum shall pass through a No. 100 U. S. Standard sieve conforming to RR-S-366.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection**. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot size**. For the purposes of sampling, a lot of asphaltum shall be no greater than 5000 pounds.

4.3 **Sampling**. Five containers shall be selected for sampling. If the lot consists of less than five containers, all containers shall be used. A 1/2-pound representative sample shall be taken from each container chosen and placed in a tightly stoppered bottle. The samples shall be thoroughly mixed to form a composite sample representative of the lot. The bottle shall be labeled to identify the sample with the lot represented.

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4.4 **Test methods.** Tests shall be performed in accordance with the following methods.

4.4.1 **Specific gravity.** Determine the specific gravity in accordance with the procedure outlined in method 23 of SS-R-406.

4.4.2 **Softening point.** Determine the softening point by the ring and ball method in accordance with method 22 of SS-R-406, using the procedure for bituminous materials having a softening point above 80° centigrade (C).

4.4.3 **Mineral matter.** Transfer a portion of approximately 10 grams of the sample to a tared porcelain crucible and weigh. Ignite and ash the material by means of a gas burner or muffle furnace. Cool the crucible in a desiccator, weigh, and calculate the weight of ash to a percentage basis.

4.4.4 **Grit.** Subject the ash obtained in 4.4.3 to tactile examination noting the presence of coarse or gritty particles.

4.4.5 **Volatile matter.** Transfer a portion of approximately 50 grams of the sample to a tared 250-milliliter (ml) beaker and weigh. Dry the beaker and contents at 100° C for 5 hours, cool in a desiccator, and weigh. Calculate the loss in weight as percentage of volatile matter.

4.4.6 **Reaction of water extract.** Transfer a weighed 10-gram portion of the sample to a 250-ml beaker, and add 100 ml of distilled water. Boil the mixture gently for a minute, cool, and decant the supernatant liquid through a paper filter. Wash the residue with four 25-ml portions of distilled water and pass each of these through the filter, combining the filtrate and wash waters. Add 3 drops of methyl orange indicator and note if the solution is of alkaline or acid reaction.

4.4.7 **Solubility.** Transfer weighed 5-gram portions of the sample to 250-ml beakers. Add 100 ml of carbon tetrachloride to one, 100 ml of petroleum ether to another, and 100 ml of carbon disulfide to the third. Allow to stand for 15 minutes or more, with occasional stirring. After settling for a few minutes, filter each solution through a tared filtering crucible containing an asbestos mat which transmits a little light. Treat any sediment in the beaker with more solvent and decant through the crucible. Finally transfer any sediment in the beaker to the crucible and wash with solvent until the washings are colorless. Air-dry the crucible until solvent-free, dry at 100° C for 1 hour, cool in a desiccator, and weigh.

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Calculate the weight of insoluble matter to a percentage basis and subtract this from 100 percent in order to obtain the percentage solubility in the solvent.

4.4.8 **Granulation**. Place an accurately weighed portion of approximately 50 grams of the sample and a silver quarter or similar object on the specified sieve assembled with a bottom pan. Cover and shake for 10 minutes by hand or by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portion passed by the sieve, and calculate the result to a percentage basis as required.

4.5 **Rejection and resubmission**. If the composite sample fails to conform to any of the requirements of this specification, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on each container at no expense to the Government. The contractor may then remove defective portions of the lot and resubmit the lot for acceptance.

5. PREPARATION FOR DELIVERY

5.1 **Packing**. Unless otherwise specified in the contract or order, asphaltum shall be packed in commercial containers so constructed as to ensure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

5.2 **Marking**. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use**. Asphaltum covered by this specification is intended for use in the manufacture of pyrotechnic and ignition compositions.

6.2 **Ordering data**. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Quantity
(c) Packing, if other than 5.1
(d) Special marking (see 5.2).

Preparing activity:
Navy-OS
(Project No. 9620-N017)

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DEPARTMENT OF THE NAVY

Naval Ordnance Station
Indian Head, MD 20640

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DOD 316

Commanding Officer
Naval Ordnance Station
Attn: Standardization Division (502)
Indian Head, Maryland 20640

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-A-356A(OS), Asphaltum (Gilsonite)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF TI E SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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